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/*TITLE: ACTIVITY 2: CASE STRUCTURES - PUSH BUTTON
  DESCRIPTION: USE THE PUSH BUTTON TO DISPLAY LETTERS AND NUMBERS ON THE SEVEN SEGMENT DISPLAY
  DATE STARTED: 12/10/18
  FINISHED: 12/20/18
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*/

const int SLA = 6, SLB = 7, SLC = 8, SLD = 9, SLE = 10, SLF = 11, SLG = 12;
int led1 = 2, led2 = 3, led3 = 4, pushbutton = 13, pushbutton2 = 5, pushbuttonvalue, pushbuttonvalue2, counter = 0, scannertime = 50, segtime = 500;
void setup()
{
  Initialize();
}

void loop()
{
  Start();
}
void Initialize()
{
  pinMode(SLA, OUTPUT);
  pinMode(SLB, OUTPUT);
  pinMode(SLC, OUTPUT);
  pinMode(SLD, OUTPUT);
  pinMode(SLE, OUTPUT);
  pinMode(SLF, OUTPUT);
  pinMode(SLG, OUTPUT);
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
  pinMode(led3, OUTPUT);

  pinMode(pushbutton, INPUT);
  pinMode(pushbutton2, INPUT);

  Serial.begin(9600);
  Serial.flush();

}


void Start()
{
  pushbuttonvalue = digitalRead(pushbutton);
  Serial.println(pushbuttonvalue);
  CheckValue();

}
```

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}

void CheckValue()
{
  switch (pushbuttonvalue)
  {

    case 1:
      {
        do
        {
          SevenSeg();                                  //Starts the letter/number sequence
          delay(2000);
          pushbuttonvalue2 = digitalRead(pushbutton2);
          Serial.println(pushbuttonvalue2);
          if (pushbuttonvalue2 == 1)          //when second button is pressed, stops the sequence
          {
            delay(500);
            break;
          }

        } while (pushbuttonvalue2 != 1);
        SegmentWrite(0, 0, 0, 0, 0, 0, 0);
        digitalWrite(led1, 0);
        digitalWrite(led2, 0);
        digitalWrite(led3, 0);
        break;
      }
    case 0:
      {
        SegmentWrite(0, 0, 0, 0, 0, 0, 0);  //All segments off at rest
        digitalWrite(led1, 0);
        digitalWrite(led2, 0);
        digitalWrite(led3, 0);
        delay(500);
        break;
      }
    default:
      {
        break;
      }
  }
}

void SegmentWrite(int a, int b, int c, int d, int e, int f, int g)
{
  digitalWrite(SLA, a);
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```
  digitalWrite(SLA, a);
  digitalWrite(SLB, b);
  digitalWrite(SLC, c);
  digitalWrite(SLD, d);
  digitalWrite(SLE, e);
  digitalWrite(SLF, f);
  digitalWrite(SLG, g);

}
void SevenSeg()
{
  SegmentWrite(1, 1, 1, 0, 1, 1, 1); //A
  delay(segtime);
  Scanner();

  SegmentWrite(1, 0, 0, 1, 1, 1, 0); //C
  delay(segtime);
  Scanner();

  SegmentWrite(1, 0, 0, 1, 1, 1, 1); //E
  delay(segtime);
  Scanner();

  SegmentWrite(1, 0, 0, 0, 1, 1, 1); //F
  delay(segtime);
  Scanner();

  SegmentWrite(1, 1, 1, 1, 0, 1, 1); //G
  delay(segtime);
  Scanner();

  SegmentWrite(0, 1, 1, 0, 0, 0, 0); //I
  delay(segtime);
  Scanner();

  SegmentWrite(0, 1, 1, 1, 1, 0, 0); //J
  delay(segtime);
  Scanner();

  SegmentWrite(0, 0, 0, 0, 1, 1, 0); //1
  delay(segtime);
  Scanner();

  SegmentWrite(1, 1, 0, 1, 1, 0, 1); //2
  delay(segtime);
  Scanner();

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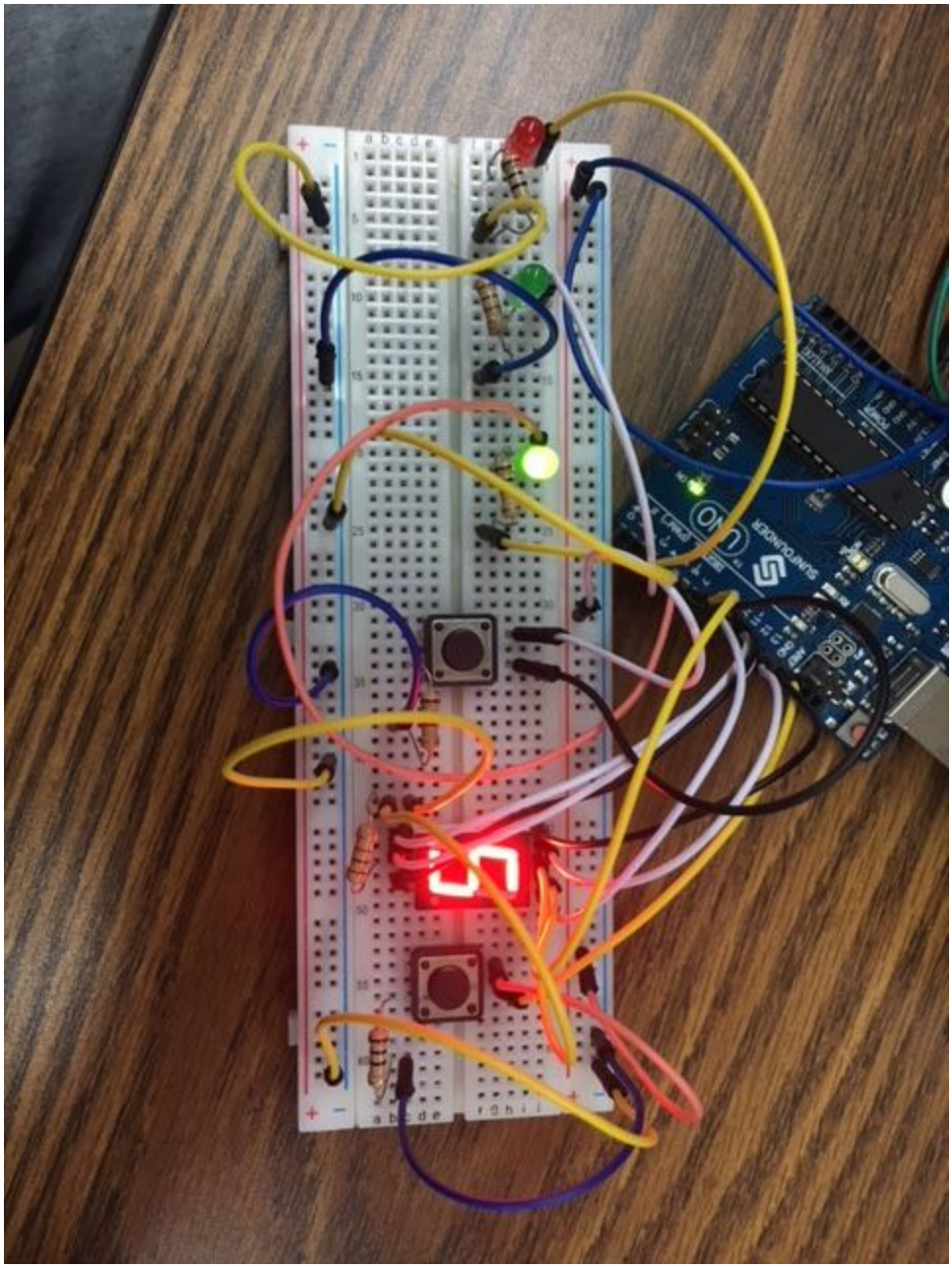
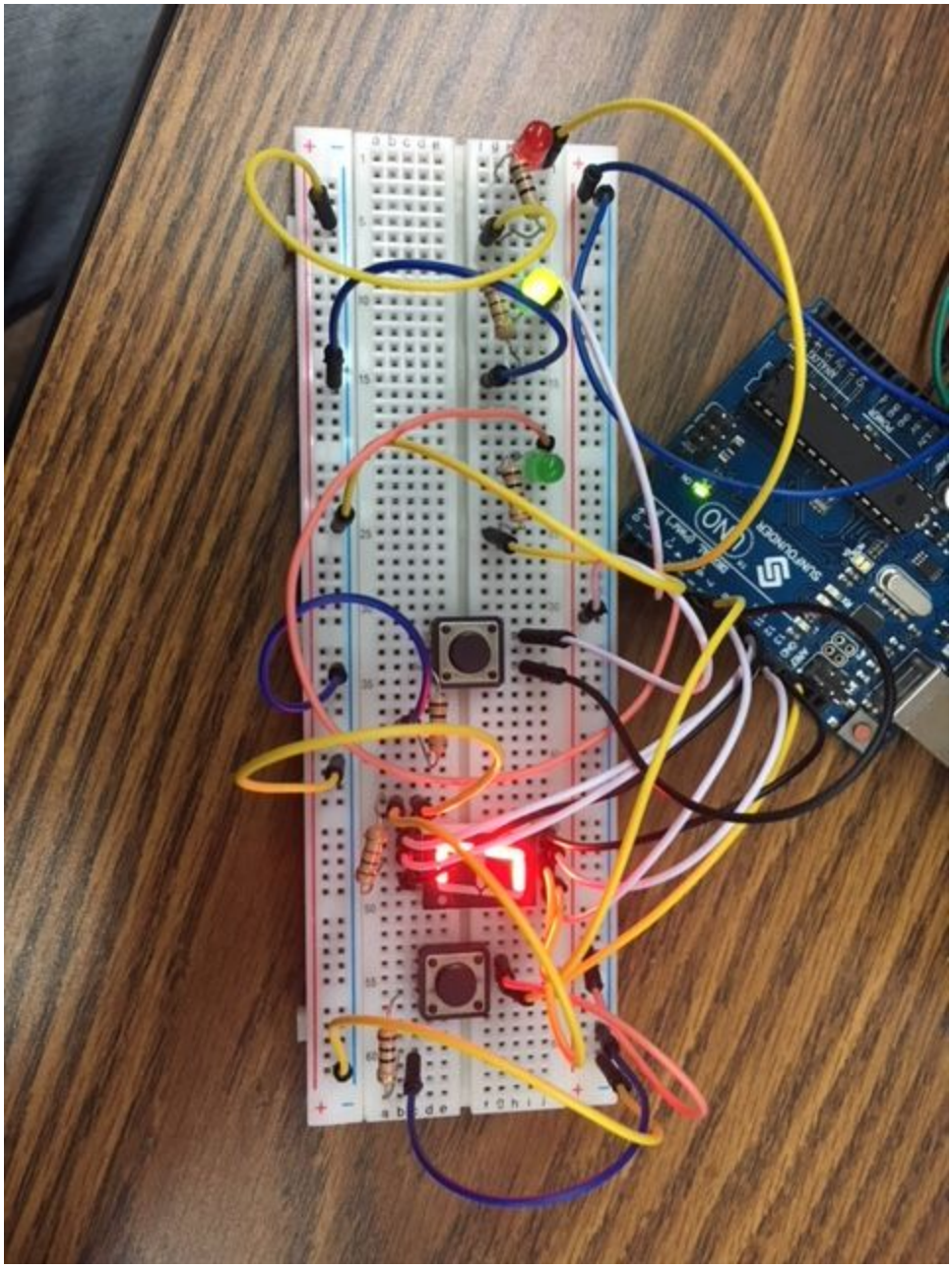
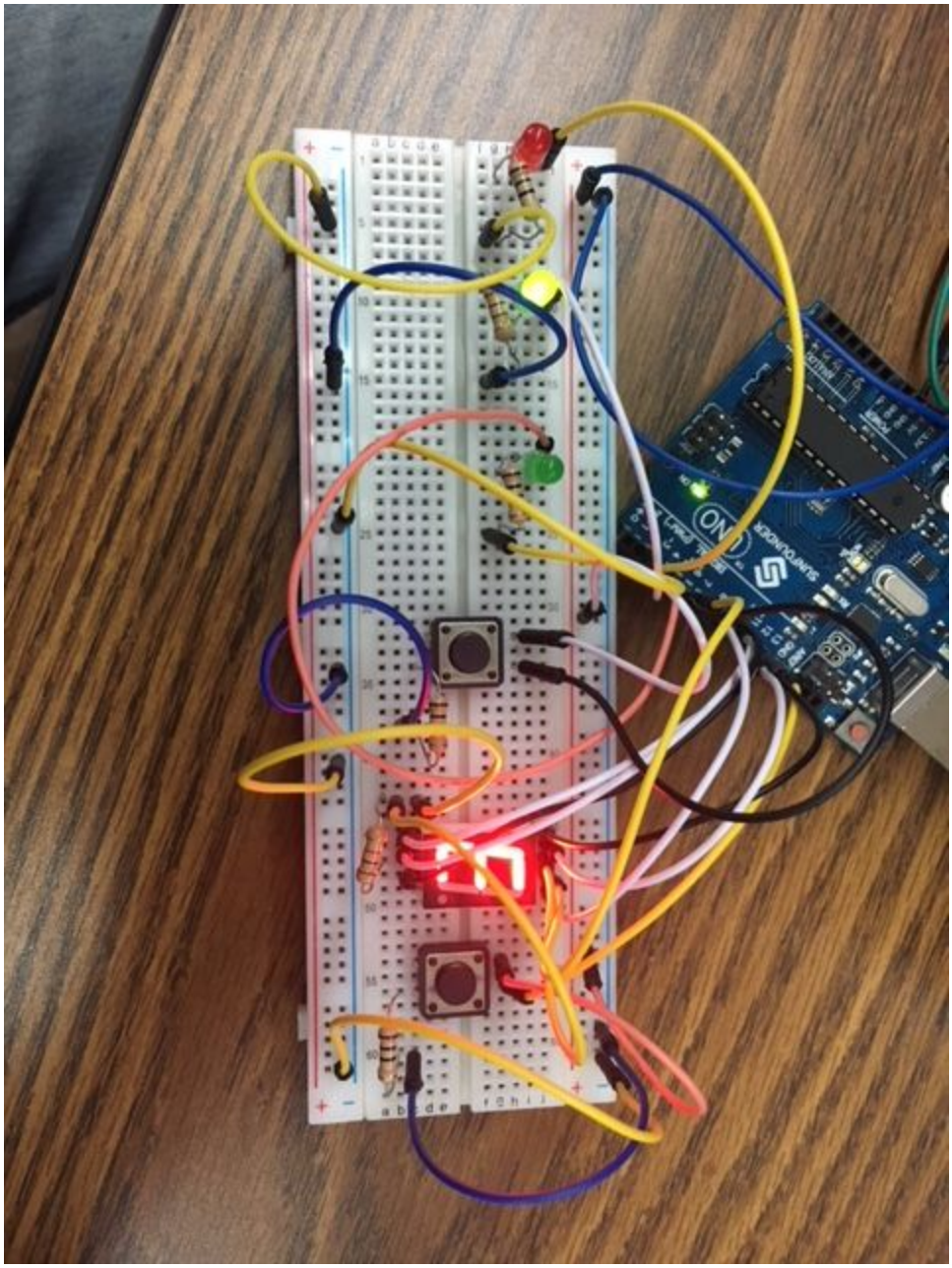
```
  SegmentWrite(0, 1, 1, 0, 0, 0, 0); //I
  delay(segtime);
  Scanner();

  SegmentWrite(0, 1, 1, 1, 1, 0, 0); //J
  delay(segtime);
  Scanner();

  SegmentWrite(0, 0, 0, 0, 1, 1, 0); //1
  delay(segtime);
  Scanner();

  SegmentWrite(1, 1, 0, 1, 1, 0, 1); //2
  delay(segtime);
  Scanner();

  SegmentWrite(1, 1, 1, 1, 0, 0, 1); //3
  delay(segtime);
  Scanner();

  SegmentWrite(0, 1, 1, 0, 0, 1, 1); //4
  delay(segtime);
  Scanner();

  SegmentWrite(1, 0, 1, 1, 0, 1, 1); //5
  delay(segtime);
  Scanner();
}
void Scanner()
{
  digitalWrite(led1, 1);  //larson scanner sequence
  digitalWrite(led2, 0);
  digitalWrite(led3, 0);
  delay(scannertime);

  digitalWrite(led1, 0);
  digitalWrite(led2, 1);
  digitalWrite(led3, 0);
  delay(scannertime);

  digitalWrite(led1, 0);
  digitalWrite(led2, 0);
  digitalWrite(led3, 1);
  delay(scannertime);
}
```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